Pairwise Preference and List Position in Fixed Hybrid Candidate Pools

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Main Test results not yet incorporated; title is provisional.

I. Introduction

Retrieval can select passages about the right topic while missing a condition that determines which passage answers a query. Negation makes this failure particularly visible: two nearly identical passages can support opposite answers [1]. After multiple retrievers contribute candidates, a reranker must both prefer the passage that satisfies the query and place it early enough to be used. These requirements motivate a joint examination of local preference and position in the returned list.

NevIR introduced contrastive passage pairs to diagnose negation sensitivity; its paired accuracy requires correct preferences for both associated queries [1]. NevIR and its reproduction already show that improvements on negation-sensitive evaluation need not translate into improvements on broader retrieval benchmarks [1], [2]. We ask a more specific question within one evaluation environment: *when the candidate set is fixed, do changes that improve the designated pairwise preference also improve the preferred passage's position among surrounding candidates?* A correct preference alone cannot answer this question: ranking the preferred passage 100th and its counterpart 101st satisfies the local ordering while leaving both far down the list.

We study this distinction in fixed pools formed by the union of dense, learned sparse, and BM25 retrieval results. All methods use the same retrieved passages, without additional retrieval or access to source text beyond those passages. We measure strict per-query preference for the officially designated preferred passage and its reciprocal rank in the full candidate list. Both metrics are conditional on pair coverage, which is reported separately. Background candidates lack relevance judgments, so these position metrics describe designated passages, not overall pool relevance or downstream answer quality.

On 371 covered confirmation queries, a feature-based ranker trained with pairwise supervision obtains 208 correct preferences versus 189 for fixed score fusion, yet has a lower mean reciprocal rank for the preferred passage (0.3169 versus 0.6017). A training variant with sampled background candidates and revised supervision restores this metric to 0.6133, but reduces correct preferences to 198. Thus, restoring position does not recover the stronger pairwise preference in this comparison. This joint intervention does not isolate the cause of recovery.

We also evaluate a bounded use of existing pointwise LLM reranking [3]. A Qwen3-14B-AWQ configuration scores query–passage inputs independently within the fusion baseline's top 20. Fusion scores break ties between discrete judgments, while passages outside the window retain their order. The evaluation concerns the combined procedure: its scoring model, bounded window, and fusion prior all contribute to the resulting order.

Among the 371 covered confirmation queries, the first Qwen reranking run yields 280 correct preferences, compared with 189 for fusion and 257 for a BGE reranker using the same window. A separate fixed-configuration rerun yields 277; both runs are retained. These are observations on a previously used confirmation set, not independent Test evidence. The study contributes a within-pool diagnosis of the separation between pairwise preference and designated-passage position, together with an evaluation of training and bounded reranking interventions against both outcomes.

II. Setting and Evaluation

Our first question (RQ1) is whether improvements in designated-pair preference coincide with improvements in designated-passage position under training interventions. The second (RQ2) concerns bounded text reranking and its cost. For each query q, we fix the deduplicated union C_q of dense, learned sparse, and BM25 results. All comparisons reuse the saved query, passage text, and route scores and ranks. Missing designated passages are not inserted. The saved route depths are 150/50/100, respectively. Dense retrieval uses `text-embedding-v4` (1,024 dimensions), learned sparse retrieval uses `doubao-embedding-vision-251215`, and BM25 uses the project's SQLite FTS5 implementation. This is an evaluation of post-retrieval ordering, without additional retrieval, passage rewriting, or answer generation.

We use the official passage preferences of NevIR [1]. The training/development/confirmation retrieval corpus contains 1,761 distinct passages from its Train and Validation material, with one passage per document. Development contains 38 pairs, including 20 previously inspected pairs and their source-group closure; confirmation contains the remaining 187 Validation pairs. Source groups connect examples through source identifiers, identical passage text, and exact or normalized query associations. Table I separates planned queries from usable directions. Of 1,871 jointly covered training directions, two with identical queries and opposing preferences are excluded, leaving 1,869 for training. Development was used for

TABLE I
DATA POPULATIONS. USED QUERIES AND SOURCE GROUPS REFER TO LOSS-ELIGIBLE TRAINING DIRECTIONS OR JOINTLY COVERED EVALUATION DIRECTIONS. COMPLETE PAIRS REQUIRE BOTH DIRECTIONS.

Partition	Planned queries	Used queries	Complete pairs	Source groups
Train	1,896	1,869	–	472
Development	76	74	37	19
Confirmation	374	371	185	96

parameter selection and early stopping. Confirmation was first used for an earlier model comparison and was subsequently reused for the present diagnostics and reranking studies; its name does not imply a fresh holdout for every intervention.

Let d_q^+ denote the officially preferred passage and d_q^- its counterpart, and let $Q_c = \{q : d_q^+, d_q^- \in C_q\}$. The primary preference measure is the fraction of queries in Q_c for which d_q^+ has a strictly higher score. For composite rerankers, the comparison uses the ordered scoring keys defined in Section III. Equal keys remain ties; deterministic passage identifiers establish list positions but never turn a score tie into a strict success. Separately, bidirectional accuracy requires both associated queries to be strictly correct and uses only complete pairs. Preferred-passage MRR is $|Q_c|^{-1} \sum_{q \in Q_c} 1/r_q^+$, where r_q^+ is its one-based position in the full pool. We also report mean ranks of both designated passages and the number of queries with both in the top 10. None of these position measures assesses the relevance of unjudged background candidates.

Official preferences define the main analysis. A later human review of development retained 19 ties and four unresolved judgments among 76 queries. Of 53 strict human preferences, 52 have joint candidate coverage: 47 agree with the official direction and five disagree. This provides a label-sensitivity analysis on already inspected data, not a replacement gold standard. Preference denominators do not shrink when judging fails; the reranker instead falls back as specified below. For window reranking, cost accounting covers all planned queries and distinguishes effective query–passage judgments, actual request attempts, tokens, and wall-clock time. Where uncertainty intervals are reported, resampling uses source groups rather than treating associated queries as independent; the list-diagnostic aggregates below are descriptive.

Test has already been used in a separate E0/background-training comparison. Its delivered aggregate underwent a technical correction that removed identifier-based tie breaking from strict correctness. The candidate inputs have since been checked, but the original per-query model scores remain missing from that delivery. Four exact Test passages shared with earlier splits were retained. At the drafting cutoff, main Qwen/BGE Test evaluation has not yielded a complete, verified comparison. We therefore use the locally reproduced confirmation evidence for the list diagnosis below; neither Test history nor the incomplete run is presented as a completed independent validation.

III. COMPARATORS AND INTERVENTIONS

Fixed fusion. The reference combines normalized dense, sparse, and BM25 scores with weights 0.70, 0.15, and 0.15. Fusion preprocessing filters route scores at 0.30/0.20/0, applies $\log(1+s)$ to sparse/BM25 scores, and min–max normalizes within each retained route; a constant-score route receives ones. Weights are renormalized over nonempty routes. Missing contributions, including passages with no surviving route score, are zero. Filtering affects fusion contributions, not membership of the saved union. These saved scores order the complete pool and supply both text rerankers' top-20 window; they are not refitted to confirmation outcomes.

Pair-trained E0. The English comparator is a LightGBM LambdaMART ranker [4] with 38 features computed from the complete candidate pool, including route scores and ranks, route agreement, and lexical matching signals. Only the two designated passages enter each training query's loss, labeled 1/0; background candidates affect pool features and inference but receive no training labels. E0 inherits the previously selected configuration with seven leaves, depth three, learning rate 0.03, and L2 regularization 1. The English adaptation uses one fit with seed 20260907, at most 300 rounds, and patience 40. Source-balanced weights and stopping on development strict preference averaged within and then across source groups yield 67 trees. Both training arms retain LambdaRank truncation level 2. Parameters were not selected by the confirmation comparison reported here.

Background training ($N = 8$). This variant retains E0's features, hyperparameters, seed, and development stopping criterion. Each training query adds eight candidates sampled without replacement from its pool outside the designated pair. The sampling seed is the first eight SHA-256 bytes of the string `20260907:`*query-id*, interpreted as a big-endian integer. Labels become 2/1/0 for the preferred passage, counterpart, and sampled background, with gains $[0, 1, 2]$. The loss grows from 3,738 to 18,690 rows over the same 1,869 queries; the fitted model has 69 trees. Background labels are weak assumptions, not human judgments of irrelevance. Sampling, label levels, and gains change together, making this a joint training intervention rather than an isolated test of background count.

Shared text-reranking procedure. Both BGE and Qwen score query–passage inputs separately within the same fusion top 20. Neither receives official pair identities, labels, or other candidates. The head is sorted by text score, then fusion score, then passage identifier; the tail keeps its fusion order. On successful scoring, strict comparisons use the key $(1, s_{\text{text}}, s_{\text{fusion}})$ in the head and $(0, 0, s_{\text{fusion}})$ in the tail. Identifier ordering is excluded from strict comparisons. Any unavailable head score restores both the original fusion order and raw fusion scores for strict comparisons, retaining that query in the evaluation. Pair-only scoring of designated passages is an auxiliary diagnostic and does not replace this candidate-window evaluation.

Scoring models. BGE uses `bge-reranker-v2-m3` [5] through the Transformers sequence-classification interface in FP16, returning raw logits without discretization. Its observed development/confirmation inputs have at most 336 tokens, with no truncation. New Qwen requests and the separate repeat use `Qwen3-14B-AWQ` in thinking mode [6], temperature 0, an 8,192-token context, and a 6,144-token output cap. The fixed prompt requests an integer score from 0 to 4: unrelated, condition contradicted or misattributed, required evidence absent, satisfied with an implicit detail, or explicitly satisfied. The prompt, model revisions, and run configurations are retained with the experimental artifacts. This integer judgment differs from a label-probability scoring rule [3].

The 14B choice preceded the later observation that an 8B configuration scored slightly higher on development top-20 reranking; we do not claim that 14B is optimal. The first Qwen confirmation run reused valid scores potentially generated under earlier output limits, allowed internal retries, and lacks its original frozen-configuration file. Additional failure-only attempts did not change its top-20 result. The separate repeat used an empty cache and no failure retries, with seven of 374 queries falling back. It measures repeat behavior on an exposed set and does not retroactively establish the first run's prospective protocol. Both scoring models are evaluated as components of the fusion-window procedure, so a combined gain alone does not establish superior isolated semantic judgment.

IV. List Diagnosis under Training Interventions

Figure 1 compares the three training/fusion conditions on the same 371 covered confirmation queries. E0 obtains 208 strict preferences (56.06%), compared with fusion's 189 (50.94%), yet its preferred-passage MRR is 0.3169 versus 0.6017. The mean ranks of the preferred passage and its counterpart are 81.91/96.27 for E0 and 3.45/3.70 for fusion. Both designated passages occur in the top 10 for only 12 queries under E0, compared with 348 under fusion. Thus, E0's higher preference count coexists with substantially deeper designated-passage positions; it is not evidence of better ordering of the full pool.

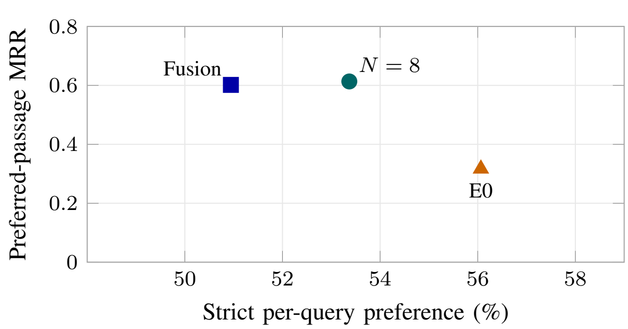

Fig. 1. Preference and preferred-passage position on the same confirmation population ($n = 371$). Higher is better on both axes. Points are observed aggregates; they do not show uncertainty intervals or establish causal mechanisms.

Background training recovers designated-passage positions while losing some of E0's strict preferences. With $N = 8$, preferred MRR reaches 0.6133, mean ranks are 2.49/2.82, and both passages appear in the top 10 for 363 queries. Strict correctness falls to 198/371, ten below E0. Relative to fusion, this variant has nine more correct preferences, but its preferred-at-1 count is one lower (135 versus 136). Development exhibits the same directional tradeoff against E0: MRR increases from 0.2851 to 0.5731 while strict correctness decreases from 44/74 to 39/74. Development was used for stopping, so this is not a second holdout validation.

The confirmation and development aggregates were reproduced locally from the saved models and candidate inputs; E0's pool scores exactly matched the earlier cache. The results are consistent with the revised supervision encouraging designated passages to outrank sampled background. They do not establish that all sampled candidates are irrelevant or identify which component of the joint intervention drives recovery. Only one background count and seed were evaluated, and no result here establishes the necessity of this intervention or its superiority over bounded text reranking.

These comparisons answer RQ1 at the level of the observed setting: the methods' ordering by strict preference differs from their ordering by designated-passage position. Both outcomes should therefore accompany the RQ2 comparison, together with separately measured costs. Planned error-category judgments and controlled rewrites can test more specific semantic hypotheses; no such explanation is inferred from these aggregate shifts alone.

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